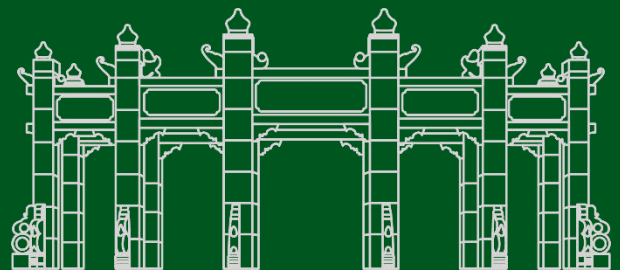
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Imaging of Nuclear Deformation with Electron Scatterings

汇报人：冯仟潞

2026.8.21







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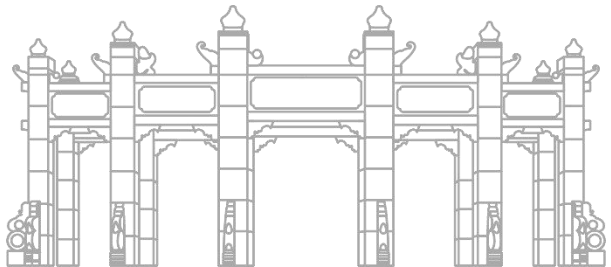
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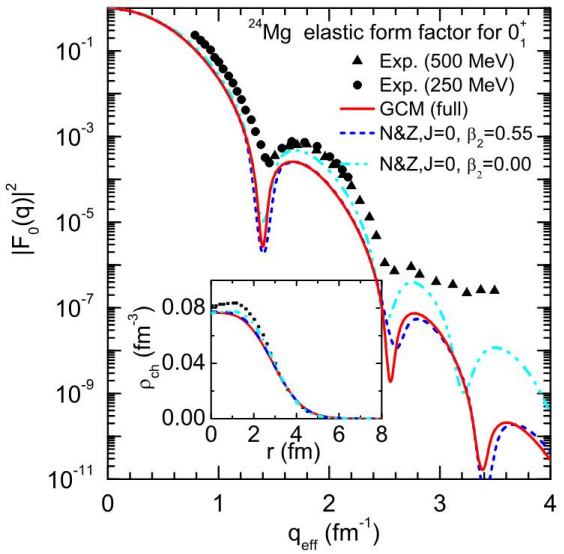
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01

Introduction



Lepton-Nucleus Scattering

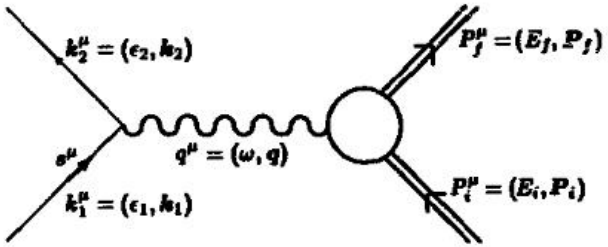


Electron-Nucleus Scattering

- charge distribution

$$\frac{d\sigma}{d\Omega} = \frac{d\sigma_{\mathrm{M}}}{d\Omega} \sum_{L\geqslant 0} |F_L(q)|^2$$

J. M. Yao., et al. PHYSICAL REVIEW C 91, 024301 (2015)

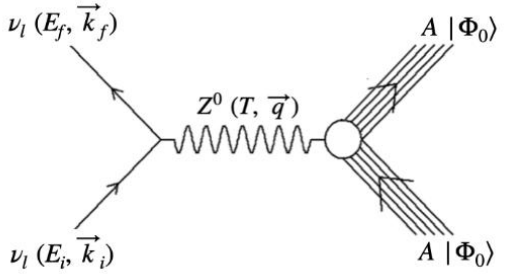


Garrido, E., et al. Nuclear Physics A 550.3 (1992): 391-420.

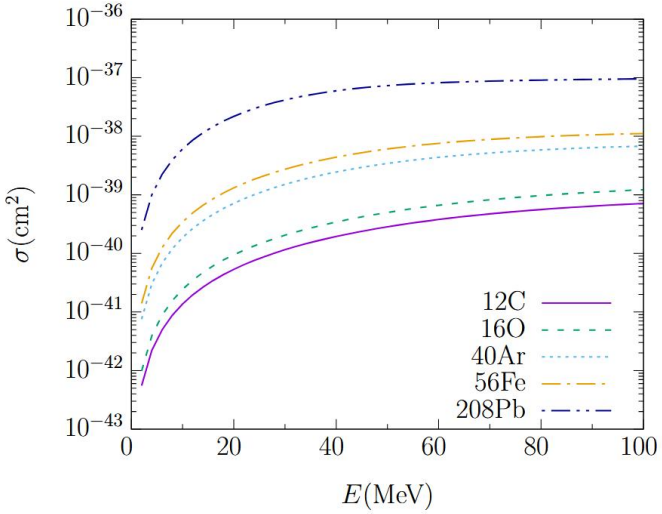
Neutrino-Nucleus Scattering

- weak charge distribution
- neutron density distribution

$$\frac{\mathrm{d}\sigma}{\mathrm{d}T} = \frac{G_F^2}{\pi} M_A \left(1 - \frac{T}{E_i} - \frac{M_A T}{2E_i^2}\right) \frac{Q_W^2}{4} F_W^2(Q^2)$$



Van Dessel, Nils, et al. Universe 9.5 (2023): 207.



Electron-Scattering

Electron Scattering of Unpolarized Nucleus:

$$\sigma_0 = V_L|F_L|^2 + V_T|F_T|^2$$

$F^{C\lambda}$: charge multipole structure

$F^{M\lambda}$: magnetic-current multipole structure

$F^{E\lambda}$: electric-current multipole structure

Electron Scattering of polarized Nucleus:

$$\sigma_{al} = \sum_{l\,even>0} \alpha_l' [P_l(\cos\theta')(-V_L F_L^l + V_T V_T^l) + P_l^2(\cos\theta')\cos 2\phi' \; V_{TT} F_{TT}^l + P_l^1(\cos\theta')\cos\phi' \; V_{TL} F_{TL}^l],$$

F_L^l: longitudinal coherent interference among Coulomb multipoles

F_T^l: transverse coherent interference among electric and magnetic multipoles

Garrido, E., et al. Nuclear Physics A 550.3 (1992): 391-420.

$F^{l=2}_{L}$ and Nuclear Deformation

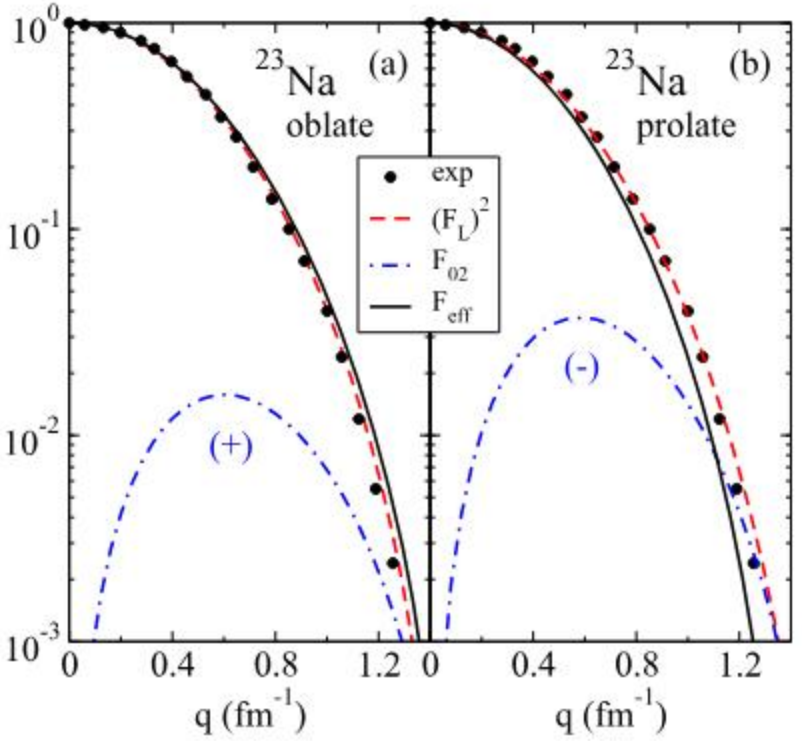
Deformation of nucleus:



spheric ($\beta_2=0$) prolate ($\beta_2>0$) oblate ($\beta_2<0$)

hard to confirm the sign of β_2

- F_{02} performs differently in prolate and oblate nucleus
- For prolate nucleus, polarization increase the cross section; for oblate nucleus, polarization behave opposite.



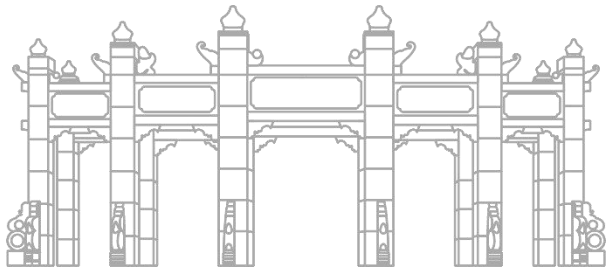
Sarriguren, Pedro. *Physical Review C* 109.2 (2024): 024312.

drawbacks of mean-field method:

- break rotational symmetry and particle-number conservation
- shape mixing is not considered

02

Theory Framework



General Cross Section (to be improved)

$$
\begin{aligned}
d\sigma &= \frac{4\alpha^2}{q^4}\left(\frac{d^3k_2}{2\varepsilon_2}\right)\frac{1}{\sqrt{(k_1\cdot p)^2}}\,\eta_{\mu\nu}W_{\mu\nu} \\
\eta_{\mu\nu} &\equiv -2\varepsilon_1\varepsilon_2\frac{1}{2}\sum_{s_1}\sum_{s_2}\bar{u}(k_1)\gamma_\nu u(k_2)\bar{u}(k_2)\gamma_\mu u(k_1) \\
W_{\mu\nu} &= (2\pi)^3\overline{\sum_i}\sum_f\langle i|\hat{J}_\nu(0)|f\rangle\langle f|\hat{J}_\mu(0)|i\rangle(\Omega E)\delta^{(4)}(p'-p+q)
\end{aligned}
$$

Walecka, John Dirk. American history 1861.1900 (1945).

Hence Lorentz invarient and current conservation, the differential cross section becomes :

$$
\frac{d^2\sigma}{d\Omega_2 d\varepsilon_2} = \sigma_M\frac{1}{M_T}[W_2(q^2,q\cdot p)+2W_1(q^2,q\cdot p)\,\tan^2\frac{\theta}{2}]
$$

In the laboratory frame, under $W_{\mu\nu} \equiv \frac{M_T^2}{E'}\delta(p_0-p_0'-q_0)w_{\mu\nu}$,there will be:

$$
2w_1(q^2) = \sum_{\lambda=\pm1}\overline{\sum_i}\sum_f\left(\frac{EE'}{M_T^2}\right)\boxed{|\langle f|\int e^{-i\mathbf{q}\cdot\mathbf{x}}\,\mathbf{e}^\dagger_{\mathbf{q}\lambda}\cdot\hat{\mathbf{J}}(\mathbf{x})\,d^3x|i\rangle|^2}
$$

$$
w_2(q^2) = \frac{q^2}{\mathbf{q}^2}w_1(q^2)+\frac{q^4}{\mathbf{q}^4}\overline{\sum_i}\sum_f\left(\frac{EE'}{M_T^2}\right)\boxed{|\langle f|\int e^{-i\mathbf{q}\cdot\mathbf{x}}\,\hat{\rho}(\mathbf{x})\,d^3x|i\rangle|^2}
$$

$$
\begin{aligned}
\hat{T}^{\rm el}_{JM}(\kappa) &\equiv \frac{1}{\kappa}\int d^3x\left[\nabla\times j_J(\kappa x)\mathscr{Y}^M_{JJ1}(\Omega_x)\right]\cdot\hat{\mathbf{J}}(\mathbf{x}) \\
\hat{T}^{\rm mag}_{JM}(\kappa) &\equiv \int d^3x\left[j_J(\kappa x)\mathscr{Y}^M_{JJ1}(\Omega_x)\right]\cdot\hat{\mathbf{J}}(\mathbf{x}) \\
\hat{M}_{JM}(\kappa) &\equiv \int d^3x\, j_J(\kappa x)Y_{JM}(\Omega_x)\,\hat{\rho}(\mathbf{x})
\end{aligned}
$$

Unpolarized and Polarized case:

For unpolarized case, $p_{M_i} = \frac{1}{2J_i+1}$, with Wigner-Eckart theorem and the orthorality , there will be

$$\frac{1}{2J_i+1}\sum_{M_i}\sum_{M_f}|\sum_{J}(-i)^J\sqrt{2J+1}\langle J_fM_f|\hat{T}_{JM}|J_iM_i\rangle|^2$$

$$=\frac{1}{2J_i+1}\sum_{JJ'}(-i)^{J-J'}\langle J_f||\hat{T}_J||J_i\rangle\langle J_f||\hat{T}_{J'}||J_i\rangle^*\sum_{M_iM_f}\langle J_fM_fJ_i-M_i|JM\rangle\langle J_fM_fJ_i-M_i|J'M\rangle$$

$$=\frac{1}{2J_i+1}\boxed{\sum_{J}|\langle J_f||\hat{T}_J||J_i\rangle|^2}$$

For polarized case,

$$\sum_{M_iM_f}p_{M_i}\langle J_iM_iJM|J_fM_f\rangle\langle J_iM_iJ'M'|J_fM_f\rangle$$

$$=(-1)^{J_f-J_i}(2J_f+1)\sqrt{(2J_i+1)}(-1)^M\sum\alpha_vW(J_iJ_iLL';vJ_f)\langle LML'-M|v0\rangle$$

there will be :

$$\sum_{M_i}\sum_{M_f}p_{M_i}|\sum_{J}(-i)^J\sqrt{2J+1}\langle J_fM_f|\hat{T}_{JM}|J_iM_i\rangle|^2$$

$$=\sum_{JJ'}(-i)^{J-J'}\langle J_f||\hat{T}_{J'}||J_i\rangle^*\langle J_f||\hat{T}_J||J_i\rangle\sum_{M_iM_f}p_{M_i}\langle J_fM_fJ_i-M_i|JM\rangle\langle J_fM_fJ_i-M_i|J'M\rangle$$

$$=(-1)^{J_f+M_f-J_i-M_i}\sqrt{(2J_i+1)(2J_f+1)}$$

$$\boxed{\sum_{JJ'}(-i)^{J-J'}\langle J_f||\hat{T}_{J'}||J_i\rangle^*\langle J_f||\hat{T}_J||J_i\rangle}\sum_{v}\alpha_vW(J_iJ_iLL';vJ_f)\langle LML'-M|v0\rangle$$

Form Factors

$$F^{\sigma\lambda} = \frac{\sqrt{4\pi}}{Z} \frac{\langle I_f \| \hat{T}^{\sigma\lambda} \| I_i \rangle}{\sqrt{2I_i+1}} \qquad (\sigma \equiv \mathrm{C, E, M})$$

Therefore,

$$\left.\frac{d\sigma}{d\Omega}\right|_{I_i^{\pi_i} \to I_f^{\pi_f}} = \sigma_{\mathrm{tot}}(\theta', \phi') = Z^2 \sigma_M f_{\mathrm{rec}}^{-1} \left[\sigma_0 + \sigma_{\mathrm{al}}(\theta', \phi')\right]$$

For unpolarized case ,

$$\sigma_0 = V_L |F_L|^2 + V_T |F_T|^2$$

For polarized case , some extra terms are added

$$\begin{aligned} \sigma_{al}(\theta', \phi') = \sum_{l=even>0} \alpha_l^{I_i} [& P_l(\cos\theta')(-V_L F_L^l + V_T F_T^l) \\ & + P_l^2(\cos\theta') \cos(2\phi') V_{TT} F_{TT}^l \\ & + P_l^1(\cos\theta') \cos\phi' V_{TL} F_{TL}^l] \end{aligned}$$

Beyond Mean-field Wavefunction

$$|JM\mu\rangle = \sum_q F^J_{\mu,q} \hat{P}^J_{M0} \hat{P}^N \hat{P}^Z |q\rangle \tag{11}$$

- Meanfield: chiral nuclear force + HFB equation

$$\begin{pmatrix} h-\lambda & \Delta \\ -\Delta^* & -h^*-\lambda \end{pmatrix} \begin{pmatrix} U_k \\ V_k \end{pmatrix} = E_k \begin{pmatrix} U_k \\ V_k \end{pmatrix} \tag{12}$$

- Projection: through AMP and PNP to keep angular momentum conservation and particle nunber conservation

$$\hat{P}^J_{MK} = \frac{\hat{J}^2}{8\pi^2} \int d\Omega \, \mathcal{D}^{J*}_{MK}(\Omega) \, \hat{R}(\Omega) \quad , \qquad \hat{P}^N = \frac{1}{2\pi} \int_0^{2\pi} d\varphi \, e^{i\varphi(\hat{N}-N)} \tag{13}$$

- GCM: shape mixing though HWG equation

$$\sum_q \left(\mathcal{H}^J_{q'q} - E^J_\mu \mathcal{N}^J_{q'q}\right) F^J_{\mu,q} = 0 \tag{14}$$

$$F^{C\lambda}(q) = \frac{\sqrt{4\pi}}{Z}\int_0^\infty drr^2\rho^{J_f\mu_f}_{J_i\mu_i,\lambda}(r)j_\lambda(qr)$$

$$\rho_L(r) = \rho^{J_f,,\mu_f}_{J_i\mu_i,L} = \sum_{nlj}\sqrt{\frac{(2j_a+1)(2l_a+1)(2L+1))}{4\pi}}R_{n_al_a}(r)R_{n_bl_b}(r)\begin{Bmatrix} l_a & 1/2 & j_a \\ j_b & L & l_b \end{Bmatrix}\sum_\lambda(J_f|||[c^\dagger_a\tilde{c}_b]_\lambda|||J_i)$$
$$\sum_{m_{ja}}(-1)^{j_b+j_a+l_b+1/2-m_{ja}}C^{\lambda,0}_{j_a,m_{ja};j_b,-m_{ja}}C^{j_b,m_{ja}}_{j_a,m_{ja};L,0}C^{l_b,0}_{l_a,0;L,0}C^{JK}_{JK\lambda0}/C^{JK}_{JKL0}$$

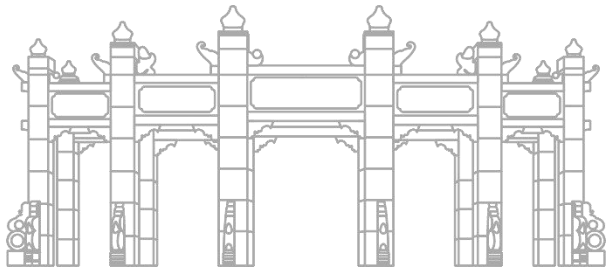
$$F^{E\lambda}_{I_i\to I_f} = \frac{q}{M}C^{I_f,K_f}_{I_i,K_i,\lambda,\mu}\sum_{ab}(-1)^{j_b-m_a}\hat{j_a}^{-1}C^{j_b,m_a}_{j_a,m_a,\lambda,\mu}(-\frac{e_i}{q}\sqrt{\frac{\lambda}{2\lambda+1}}(j_a||\mathbf{M}_{\lambda\lambda+1}(\mathbf{r}_i)\cdot\nabla||j_b)$$
$$+\frac{e_i}{q}\sqrt{\frac{\lambda+1}{2\lambda+1}}(j_a||\mathbf{M}_{JJ-1}\cdot\nabla||j_b)+\frac{\mu_i}{2}(j_a||\mathbf{M}_{JJ}(\mathbf{r}_i)\cdot\sigma(i)||j_b))$$
$$\sum_{\lambda'\mu'}C^{\lambda',\mu'}_{j_a,m_a,j_b,-m_a}C^{I_f,K_f}_{I_i,K_i,\lambda',\mu'}(I_f|||[c^\dagger_a\tilde{c}_b]_{\lambda'}|||I_i)$$

quadrupole deformation parameter β_2 → nucleon density distribution → form factors → cross section

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03

Results



Energy level for ^{21}Ne and ^{23}Na

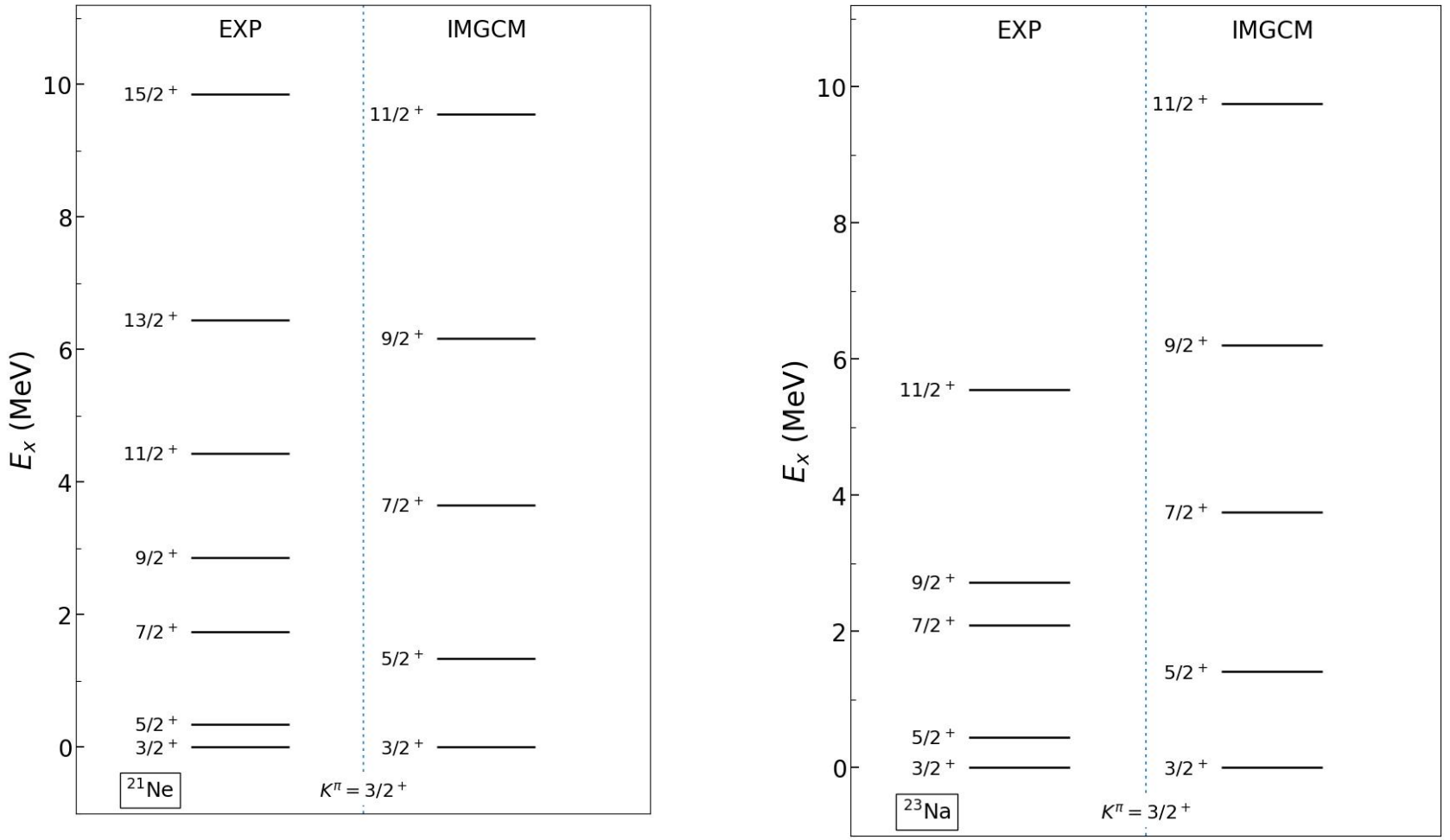


Fig1：The left panel compares the experimental energy levels of ^{21}Ne with those calculated using the IMGCM, while the right panel shows the corresponding comparison for ^{23}Na.

Longitudinal Form Factors of Different Deformations for ^{21}Ne

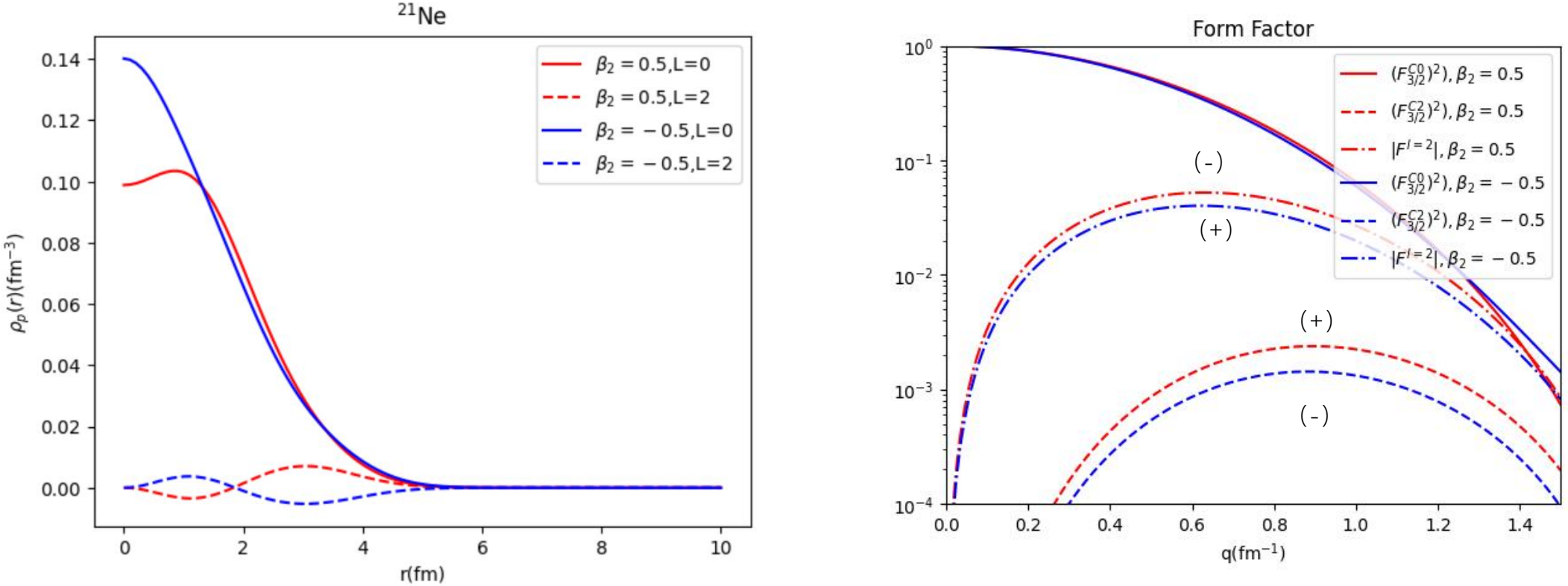


Fig 2: The left panel shows the proton density distributions of the ^{21}Ne nucleus for the monopole (l=0) and quadrupole (l=2) components in the prolate and oblate configurations, respectively. The right panel shows the squared form factors and coherent form factors of the monopole and quadrupole components for the prolate and oblate configurations.

Longitudinal Part of Different Deformations for ^{21}Ne

$$|F_{eff}|^2 = |F_L|^2 + F_{02} = |F^{C0}|^2 + |F^{C2}|^2 + F_{02}$$

$$\frac{\sigma_0 + \sigma_{al}}{\sigma_0} = \frac{|F_{eff}|^2}{|F_L|^2}$$

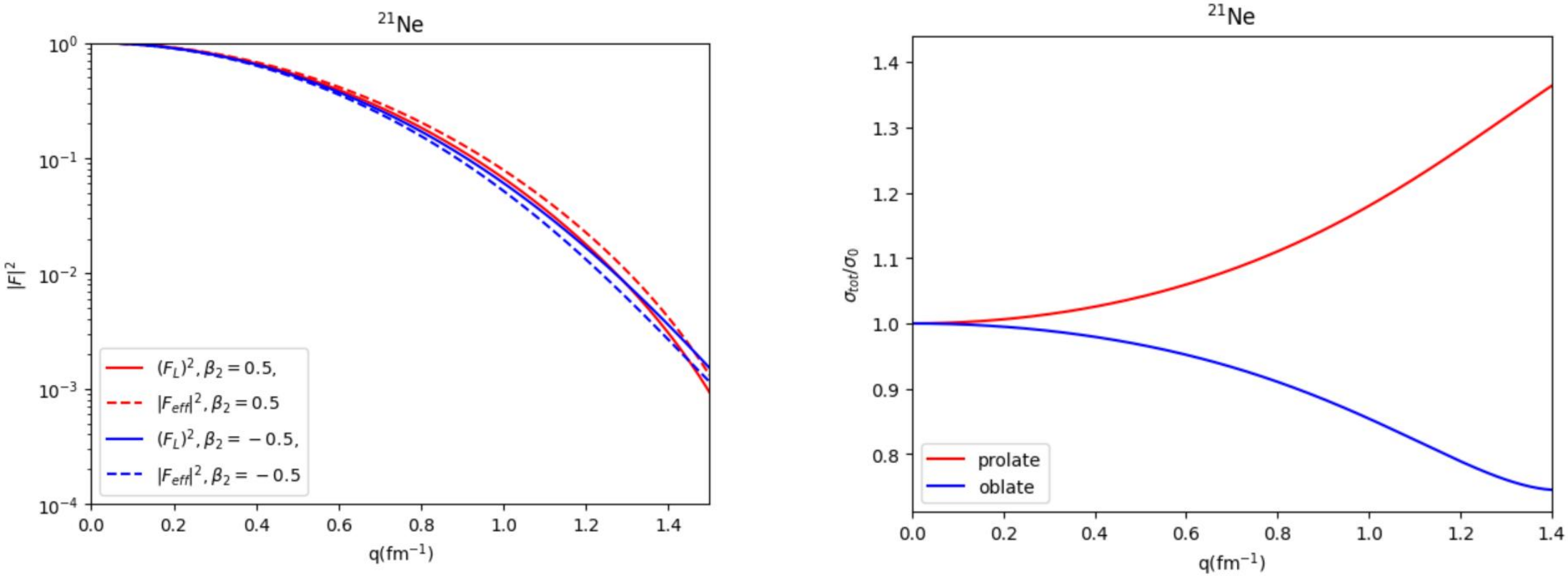


Figure 3: The left panel shows the sum of the squared form factors of different multipoles and the effective form factors for ^{21}Ne in the prolate and oblate configurations. The right panel shows the ratio of the scattering cross sections for polarized and unpolarized ^{21}Ne nuclei in the prolate and oblate configurations.

Transverse Form Factors of Different Deformations for ^{21}Ne (to be improved)

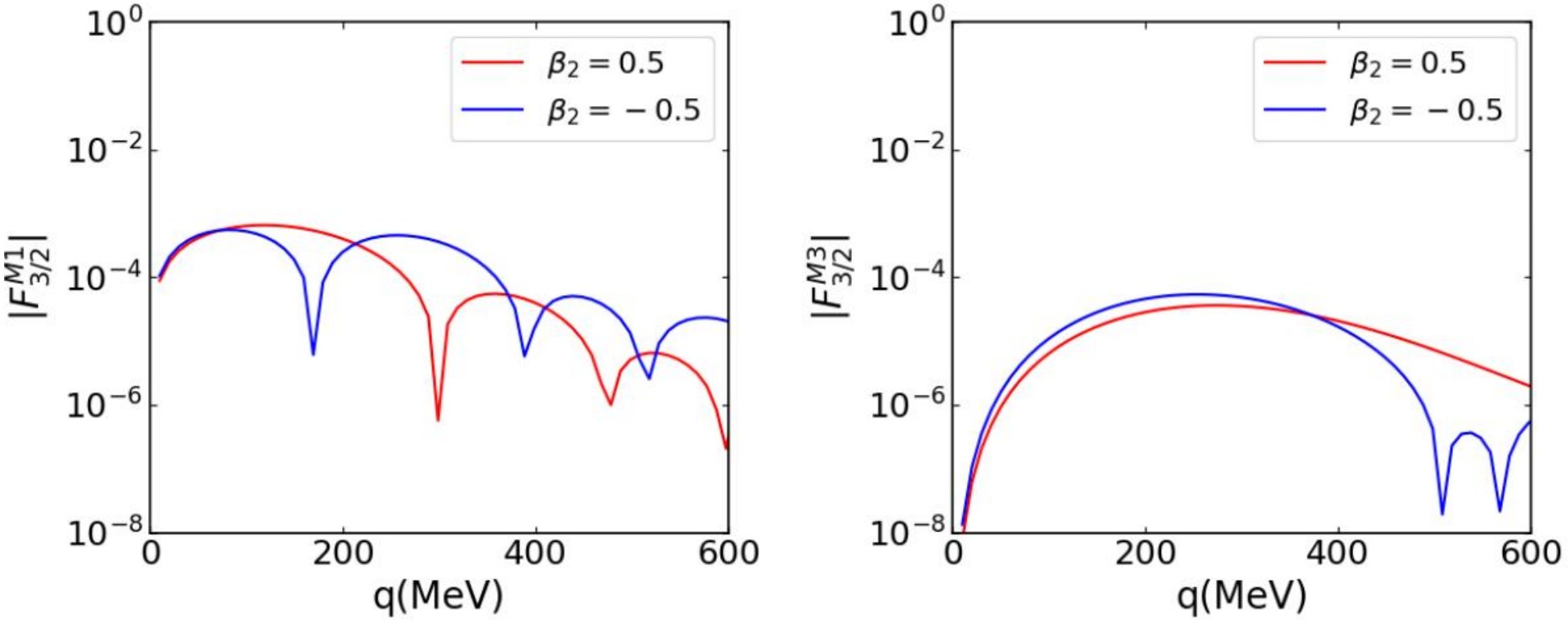


Fig 4: The left panel shows the absolute value of F^{M1}. The right panel shows the absolute value of F^{M3}.

Transverse Form Factors of Different Deformations for ^{21}Ne

$$F_T^l = \sum_{\lambda,\lambda'} C(\lambda,\lambda',I_f,I_i,l)\begin{pmatrix}\lambda & \lambda' & l\\ 1 & -1 & 0\end{pmatrix} \times (F^{M\lambda}F^{M\lambda'} + F^{E\lambda}F^{E\lambda'} + 2F^{M\lambda}F^{E\lambda'}),$$

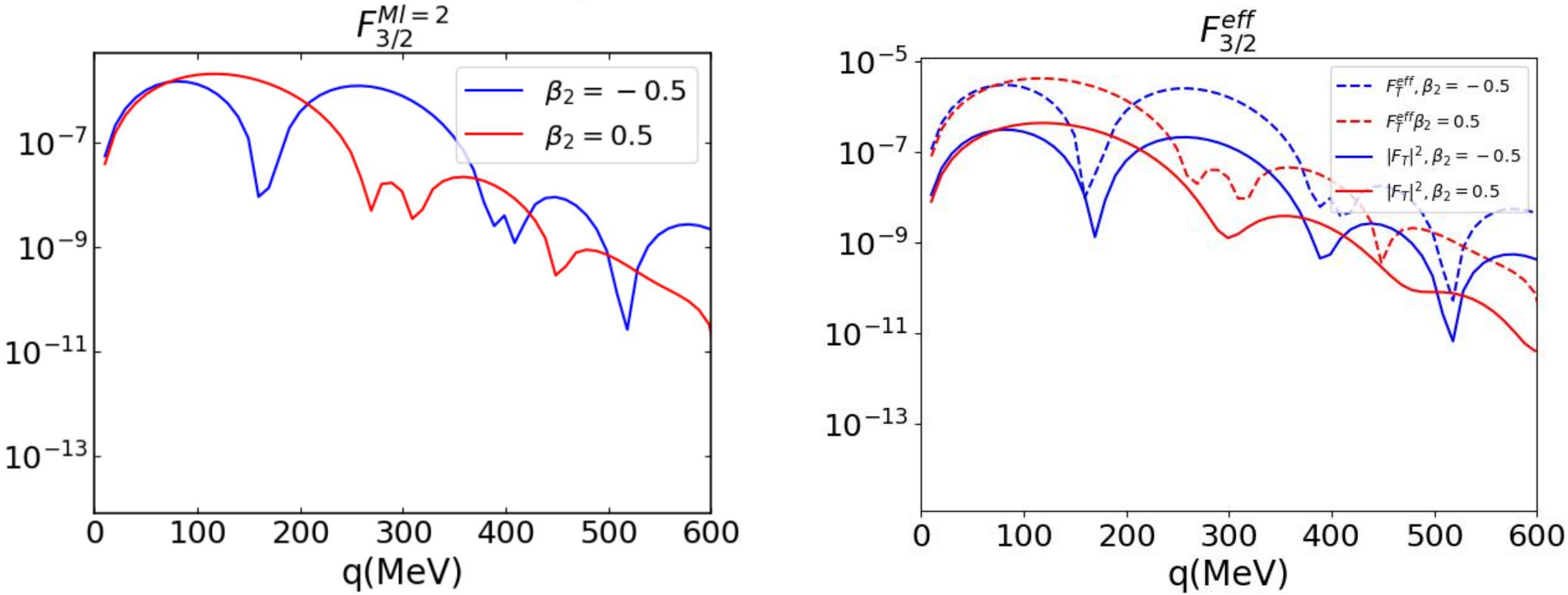


Fig 5: The left panel shows the coherent form factors. The right panel shows the effective form factors.

Comparison of MF and BYM method for ^{23}Na

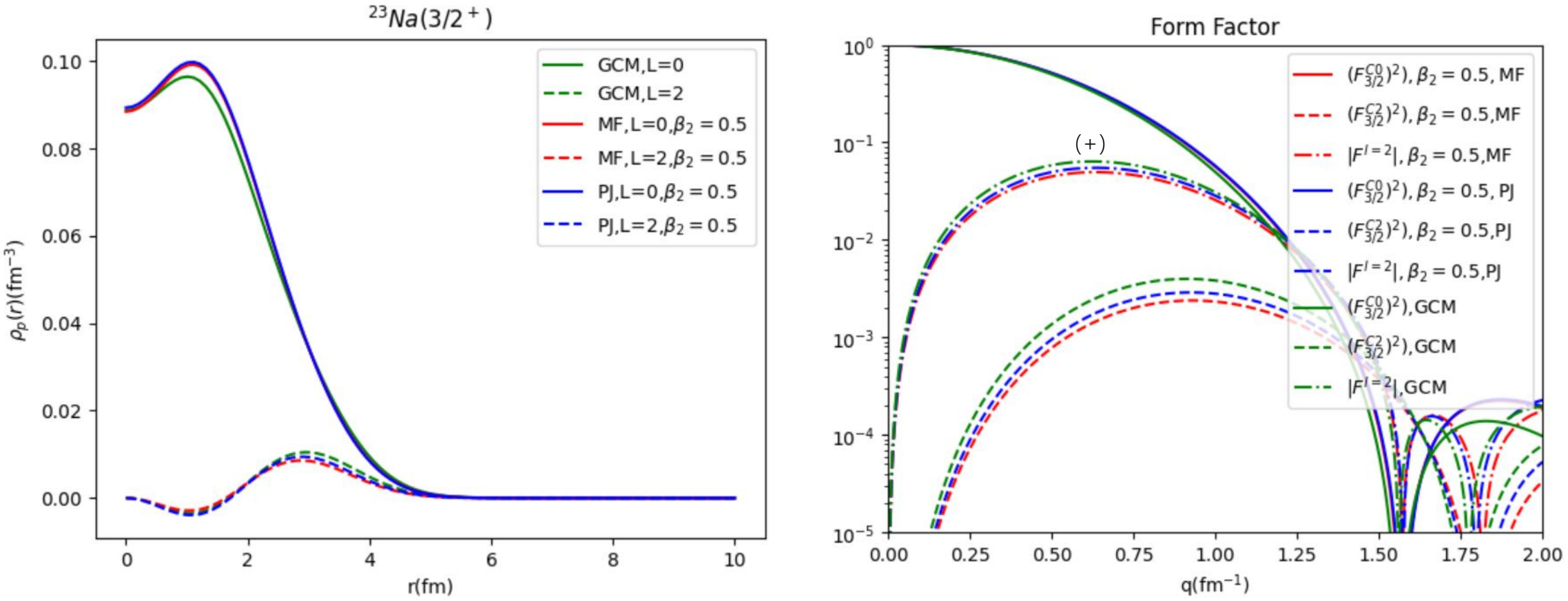
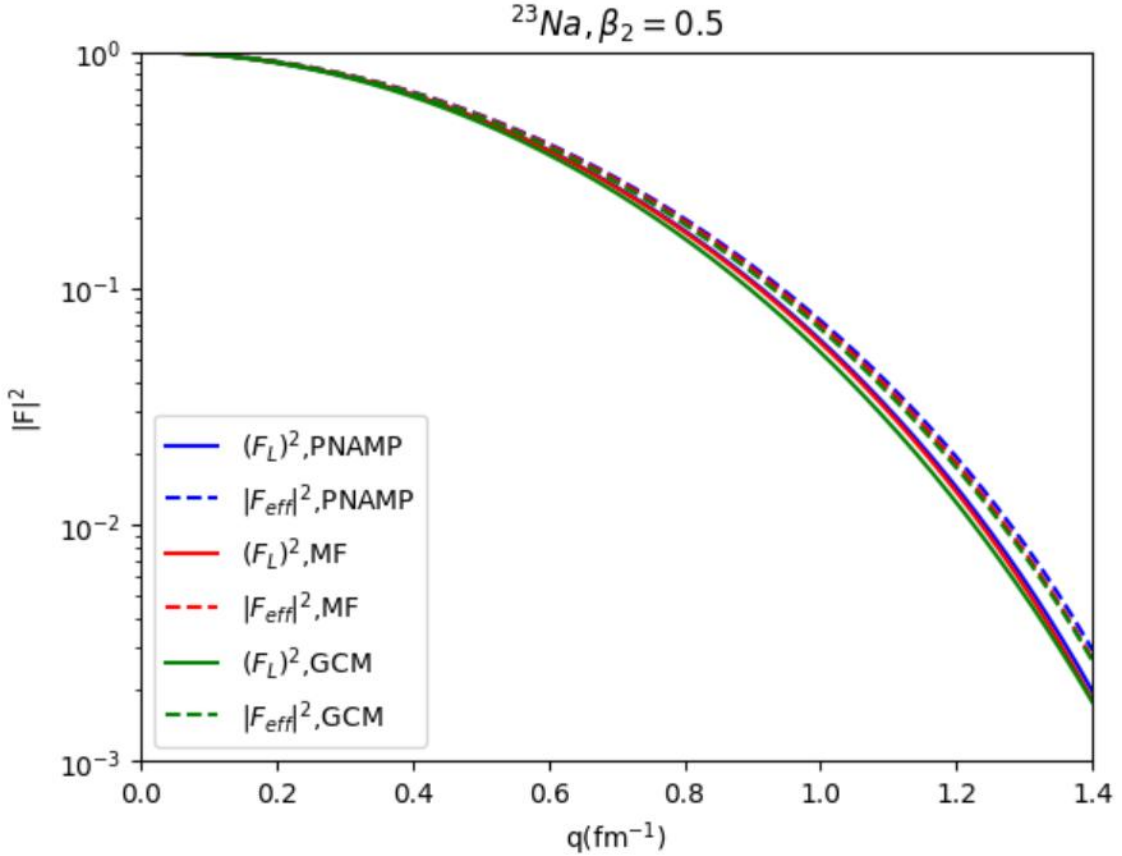


Figure 6: The left panel shows the proton density distributions of the monopole (l=0) and quadrupole (l=2) components of ^{23}Na, calculated using different wave functions. The right panel compares the squared monopole and quadrupole form factors and their corresponding coherent form factors obtained using different wave functions.

Comparison of MF and BYM method for ^{23}Na

$$|F_{eff}|^2 = |F_L|^2 + F_{02} = |F^{C0}|^2 + |F^{C2}|^2 + F_{02}$$

$$\frac{\sigma_0 + \sigma_{al}}{\sigma_0} = \frac{|F_{eff}|^2}{|F_L|^2}$$



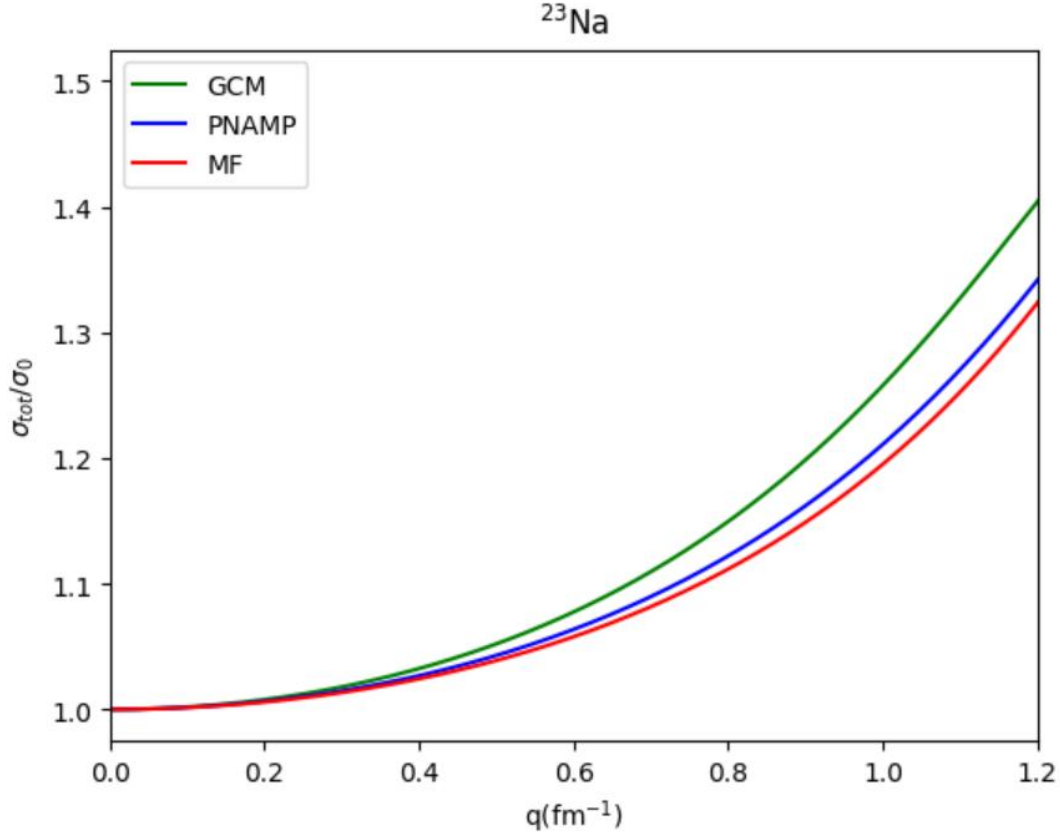


Figure 7: The left panel shows the sum of the squared form factors of different multipoles and the effective form factors of ^{23}Na obtained using different wave functions. The right panel shows the ratios of the scattering cross sections for polarized and unpolarized ^{23}Na nuclei calculated with different wave functions

Comparison of the Results from IMGCM and Experiment

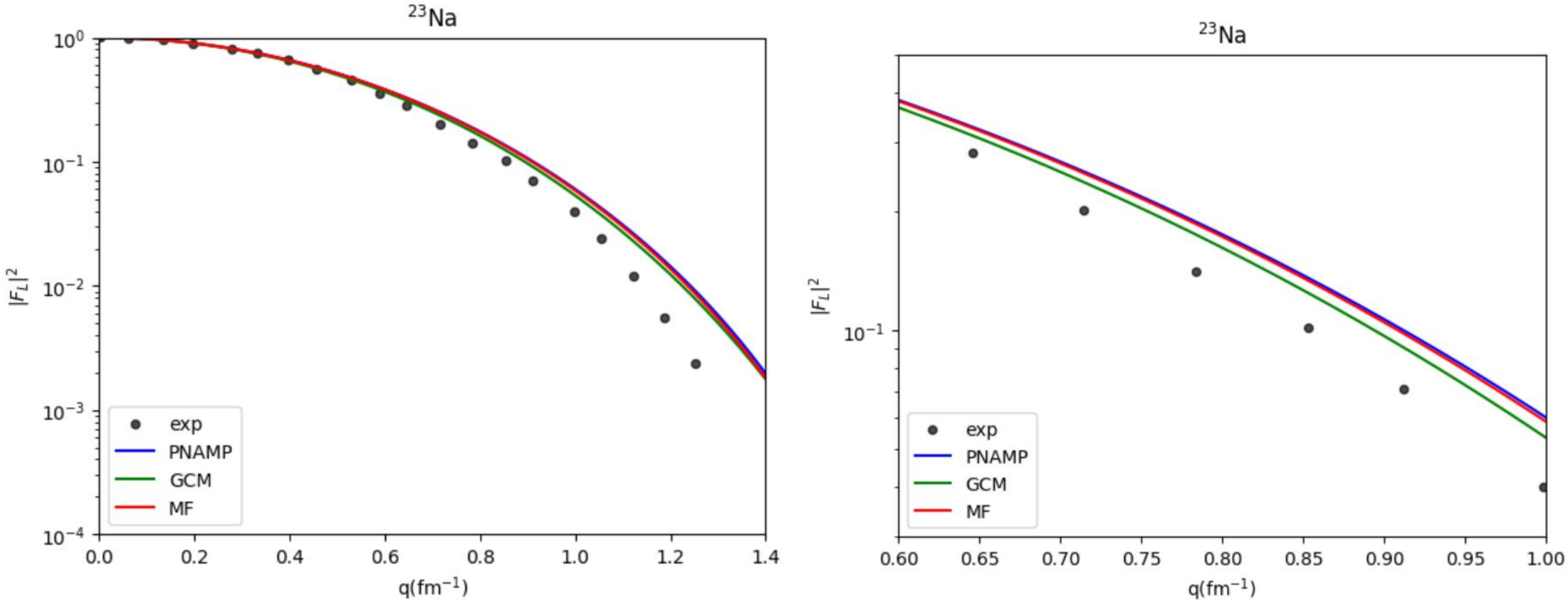
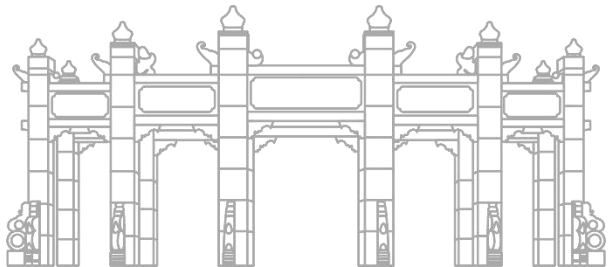


Figure 8: Comparison of the sum of the squared form factors of different multipoles for ^{23}Na, calculated using different wave functions, with experimental data.

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04

Summary



Conclusion

- The elastic scattering of electrons from a polarized odd-A nucleus is investigated using the IMGCM approach.
- The effects of nuclear deformation on the scattering cross section are analysed, revealing coherent enhancement (suppression) of the scattering cross section for prolate (oblate) deformed nuclei
- Projection and shape-mixing effects further enhance the coherent enhancement (suppression) of the scattering cross section

Outlook

- Other many-body methods can be used (MR-CDFT) to confirm the conclusion
- Other form factors from other process of scattering can be studied
- The contributions from two-body density can be considered.

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Thanks!

冯仟潞